

PAPER_1557_M_H_HIGGS_125_10

Daniel T. Murphy

$$\text{PAPER_1557} \text{ — Higgs Boson Mass} = m_H = 2 \cdot A_5 + N_{CH} - D_{\text{phys}} + F_{\text{TRZ}} \cdot \text{SSQ} + F_{\text{TRZ}}^2 \cdot D_{\text{BSFG}} + F_{\text{TRZ}}^2 \cdot \text{SSQ}^2 \approx 125.12 \text{ GeV}$$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Particle Physics

Date: June 18, 2026

Status: CLOSED — 0.016% closure from PAPER_1209HH_S656

Observation

PAPER_1209HH_S656 (Particle Physics Unified Proof Set) lists **Higgs Boson Mass** with the integer-primitive expression:

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$$\text{Higgs Boson Mass} = m_H = 2 \cdot A_5 + N_{CH} - D_{\text{phys}} + F_{\text{TRZ}} \cdot \text{SSQ} + F_{\text{TRZ}}^2 \cdot D_{\text{BSFG}} + F_{\text{TRZ}}^2 \cdot \text{SSQ}^2 \approx 125.12 \text{ GeV}$$

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Status: **0.016%**.

UQFF Closed Identity

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$$m_H = 2 \cdot A_5 + N_{CH} - D_{\text{phys}} + F_{\text{TRZ}} \cdot \text{SSQ} + F_{\text{TRZ}}^2 \cdot D_{\text{BSFG}} + F_{\text{TRZ}}^2 \cdot \text{SSQ}^2 \approx 125.12 \text{ GeV } 0.016\%$$

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Physical Interpretation

Clean integer-primitive form at first order: $2 \cdot A_5 + N_{CH} - D_{\text{phys}} = 120 + 9 - 4 = 125 \text{ GeV}$ — exactly the Higgs boson mass to 3 significant figures, before small F_{TRZ} corrections refine to 125.12 vs CODATA 125.10.

NOT REPLACEMENT

CODATA/PDG treats Higgs Boson Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S656
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_h_higgs_125_10"})`

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